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function: a rule that assigns to a unique value $f(x) \in \text{Range}$ to every $x \in \text{Domain}$

domain: 定義域, input 範圍

range: 值域, output 範圍

integer floor function $\lfloor x \rfloor$ $\lfloor 0.5 \rfloor = 0$ $\lfloor 1.3 \rfloor = 1$

integer ceiling function $\lceil x \rceil$ $\lceil 0.5 \rceil = 1$ $\lceil 1.3 \rceil = 2$

monotonic function: 單調函數

even function: 偶函數, for $x \in D$ $f(x) = f(-x)$

odd function: 奇函數, for $x \in D$ $f(x) = -f(-x)$

power function: 冪函數: x^a $a \in \mathbb{R}$

constant function: 常數函數

linear function: 線性函數

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independent number:

difference quotient: 差商: $\frac{\Delta h}{h} = a$

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coordinate vs. axis

x-coordinate x座標

x-axis x軸

y-coordinate y座標

y-axis y軸

xy plane xy平面

parabola 拋物線

piecewise function: 分段函数: a function that is defined by different formulas in different domains.

absolute value: The distance from a number to 0 on the real number line, it is always positive or 0.

symmetry: 對稱性